

# HDOC- 23

## Question:

Write a C program that displays the following menu and uses a switch statement to print the basic square/rectangular pattern selected by the user. Use nested loops without arrays. Each pattern may be implemented in a separate parameterless function with no return value.

## Task1: Algorithm:

**Step 1:** Start.

**Step 2:** Declare an integer variable choice.

**Step 3:** Display the menu:

1. Solid Square Pattern
2. Solid Rectangular Pattern
3. Row-Number Pattern
4. Column-Number Pattern
5. Exit

**Step 4:** Input choice.

**Step 5:** Use a switch statement:

- If choice = 1, call solidSquare().
- If choice = 2, call solidRectangle().
- If choice = 3, call rowNumber().
- If choice = 4, call columnNumber().
- If choice = 5, exit.
- Otherwise, display "Invalid choice".

**Step 6:** Repeat the menu until the user selects Exit.

**Step 7:** Stop.

## 2. Algorithm for Each Function

### Function 1: Solid Square Pattern

The question requires an  $n \times n$  square of stars.

1. Start function.
2. Input n.
3. Check whether  $n > 0$ .

4. Repeat i from 1 to n.
5. For each row, repeat j from 1 to n.
6. Print \*.
7. After completing one row, print a new line.
8. Return to main().

### **Function 2: Solid Rectangular Pattern**

The rectangle takes m rows and n columns.

1. Start function.
2. Input m and n.
3. Check whether both are positive.
4. Repeat i from 1 to m.
5. Repeat j from 1 to n.
6. Print \*.
7. Print a new line after every row.
8. Return to main().

### **Function 3: Row-Number Pattern**

Every value in a row must be equal to the row number.

1. Input m and n.
2. Repeat i from 1 to m.
3. Repeat j from 1 to n.
4. Print i.
5. Print a new line after every row.
6. Return.

### **Function 4: Column-Number Pattern**

Every value in a column must be equal to the column number.

1. Input m and n.
2. Repeat i from 1 to m.
3. Repeat j from 1 to n.
4. Print j.
5. Print a new line after every row.
6. Return.

**Task2: Test case table:**

| Test Case | Input                     | Expected Output                         | Actual Output                 | Result |
|-----------|---------------------------|-----------------------------------------|-------------------------------|--------|
| 1         | Choice = 1, n = 4         | 4 × 4 solid square                      | 4 × 4 solid square            | Pass   |
| 2         | Choice = 1, n = 1         | Single *                                | Single *                      | Pass   |
| 3         | Choice = 1, n = 0         | Invalid input                           | Invalid input                 | Pass   |
| 4         | Choice = 2, m = 3, n = 5  | 3 × 5 solid rectangle                   | 3 × 5 solid rectangle         | Pass   |
| 5         | Choice = 2, m = 2, n = 3  | 2 × 3 solid rectangle                   | 2 × 3 solid rectangle         | Pass   |
| 6         | Choice = 2, m = -2, n = 4 | Invalid input                           | Invalid input                 | Pass   |
| 7         | Choice = 3, m = 4, n = 4  | Rows containing 1, 2, 3, 4 respectively | Correct row-number pattern    | Pass   |
| 8         | Choice = 3, m = 3, n = 2  | 1 1, 2 2, 3 3                           | Correct pattern               | Pass   |
| 9         | Choice = 4, m = 4, n = 4  | Each row: 1 2 3 4                       | Correct column-number pattern | Pass   |
| 10        | Choice = 4, m = 2, n = 5  | Each row: 1 2 3 4 5                     | Correct pattern               | Pass   |
| 11        | Choice = 7                | Invalid choice                          | Invalid choice                | Pass   |
| 12        | Choice = 5                | Program terminates                      | Program terminates            | Pass   |

## Task3: C program:

```
#include <stdio.h>
void solidSquare()
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    if (n <= 0)
    {
        printf("Invalid input\n");
        return;
    }
    for (i = 1; i <= n; i++)
    {
        for (j = 1; j <= n; j++)
        {
            printf("* ");
        }
        printf("\n");
    }
}

void solidRectangle()
{
    int m, n, i, j;
    printf("Enter number of rows and columns: ");
    scanf("%d %d", &m, &n);
    if (m <= 0 || n <= 0)
    {
        printf("Invalid input\n");
        return;
    }
    for (i = 1; i <= m; i++)
    {
        for (j = 1; j <= n; j++)
        {
            printf("* ");
        }
        printf("\n");
    }
}

void rowNumber()
{
    int m, n, i, j;
    printf("Enter number of rows and columns: ");
```

```

scanf("%d %d", &m, &n);
if (m <= 0 || n <= 0)
{
    printf("Invalid input\n");
    return;
}
for (i = 1; i <= m; i++)
{
    for (j = 1; j <= n; j++)
    {
        printf("%d ", i);
    }
    printf("\n");
}
}

void columnNumber()
{
    int m, n, i, j;
    printf("Enter number of rows and columns: ");
    scanf("%d %d", &m, &n);
    if (m <= 0 || n <= 0)
    {
        printf("Invalid input\n");
        return;
    }
    for (i = 1; i <= m; i++)
    {
        for (j = 1; j <= n; j++)
        {
            printf("%d ", j);
        }
        printf("\n");
    }
}

int main()
{
    int choice;
    do
    {
        printf("\n----- PATTERN MENU ----- \n");
        printf("1. Solid Square Pattern\n");
        printf("2. Solid Rectangular Pattern\n");
        printf("3. Row-Number Pattern\n");
        printf("4. Column-Number Pattern\n");
        printf("5. Exit\n");

        printf("Enter your choice: ");
    }
    while (choice != 5);
}

```

```
scanf("%d", &choice);
switch (choice)
{
    case 1:
        solidSquare();
        break;
    case 2:
        solidRectangle();
        break;
    case 3:
        rowNumber();
        break;
    case 4:
        columnNumber();
        break;
    case 5:
        printf("Program terminated.\n");
        break;
    default:
        printf("Invalid choice. Please try again.\n");
}
} while (choice != 5);
return 0;
}
```

## Task4:Output:

```
(kali㉿kali)-[~/hdoc23.c]
$ ./a.out

——Pattern menu——
1. Solid Square Pattern
2. Solid recatngular pattern
3. Row-Number Pattern
4. column-number pattern
5. Exit
Enter your choice: 1
Enter number of rows: 4
****
****
****
****

——Pattern menu——
1. Solid Square Pattern
2. Solid recatngular pattern
3. Row-Number Pattern
4. column-number pattern
5. Exit
Enter your choice: 1
Enter number of rows: 1
*

——Pattern menu——
1. Solid Square Pattern
2. Solid recatngular pattern
3. Row-Number Pattern
4. column-number pattern
5. Exit
Enter your choice: 1
Enter number of rows: 0
Invalid input

——Pattern menu——
1. Solid Square Pattern
2. Solid recatngular pattern
3. Row-Number Pattern
4. column-number pattern
5. Exit
Enter your choice: 2
Enter number of rows and columns: 3
5
*****
*****
*****
```

——Pattern menu——

1. Solid Square Pattern
2. Solid recatngular pattern
3. Row-Number Pattern
4. column-number pattern
5. Exit

Enter your choice: 2

Enter number of rows and columns: 2

3

\*\*\*

\*\*\*

——Pattern menu——

1. Solid Square Pattern
2. Solid recatngular pattern
3. Row-Number Pattern
4. column-number pattern
5. Exit

Enter your choice: 2

Enter number of rows and columns: -2

4

Invalid input

——Pattern menu——

1. Solid Square Pattern
2. Solid recatngular pattern
3. Row-Number Pattern
4. column-number pattern
5. Exit

Enter your choice: 3

Enter number of rows and columns: 4

4

1 1 1 1

2 2 2 2

3 3 3 3

4 4 4 4

——Pattern menu——

1. Solid Square Pattern
2. Solid recatngular pattern
3. Row-Number Pattern
4. column-number pattern
5. Exit

Enter your choice: 3

Enter number of rows and columns: 3

2

1 1

2 2

3 3

——Pattern menu——

1. Solid Square Pattern
2. Solid recatngular pattern
3. Row-Number Pattern
4. column-number pattern
5. Exit

Enter your choice: 4

Enter number of rows and column: 4

4

1234

——Pattern menu——

1. Solid Square Pattern
2. Solid recatngular pattern
3. Row-Number Pattern
4. column-number pattern
5. Exit

Enter your choice: 4

Enter number of rows and column: 2

5

12345

——Pattern menu——

1. Solid Square Pattern
2. Solid recatngular pattern
3. Row-Number Pattern
4. column-number pattern
5. Exit

Enter your choice: 7

Invalid choice. Please try again.

——Pattern menu——

1. Solid Square Pattern
2. Solid recatngular pattern
3. Row-Number Pattern
4. column-number pattern
5. Exit

Enter your choice: 5

Program terminated.